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Development and Validation of the New Somatomorphic Matrix–Male: A Figural Rating Scale for Measuring Male Actual–Ideal Body Discrepancy

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This study aimed to develop and examine the validity evidence of the New Somatomorphic Matrix–Male (NSM-M) as a measure of actual–ideal body discrepancy. The NSM-M is a visual matrix of 34 male bodies that vary along two dimensions: body fat percentage (range = 4–40%) and muscularity (fat-free mass index; range = 16.5–30 kg/m²; 36.4–66.1 lbs/m²). In Study 1, participants were 2,733 sexual minority males recruited via a geosocial networking smartphone application. Participants selected their “actual” and “ideal” bodies on the NSM-M and additionally completed questionnaires measuring body dissatisfaction, eating disorder symptoms, quality of life, and anabolic steroid use. In Study 2, participants were 33 males from a community sample. They completed the NSM-M across two sessions, 7–14 days apart. NSM-M scores demonstrated concurrent and convergent validity as a measure of male body dissatisfaction. In addition, NSM-M scores discriminated between steroid users and nonusers, and between body mass index categories. The NSM-M also demonstrated sound test–retest reliability across a 7- to 14-day period. In sum, the NSM-M is a validated figural measure of male actual–ideal body discrepancy with notable strengths, including fast administration time and enhanced ecological validity.

Public Significance Statement

The NSM-M is a new valid and reliable tool that returns fast, sensitive measures of body image and body dissatisfaction for men. The NSM-M could be used to aid diagnosis and treatment progression of body dissatisfaction-related disorders such as eating disorders and muscle dysmorphia.

Keywords: measurement, figural rating scale, matrix, body image, men

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Body dissatisfaction can be defined as negative subjective evaluation of one’s own body as a whole, or specific aspects of one’s own body, such as body shape, size, muscle tone, or weight (Grogan, 2016). Body dissatisfaction has been linked to eating disorder symptoms (Hospers & Jansen, 2005; Mayo & George, 2014; Olivardia, Pope, Borowiecki, & Cohane, 2004), depressive symptomatology (Olivardia et al., 2004; Paxton, Neumark-Sztainer, Hannan, & Eisenberg, 2006), obesity (Mond, van den

Berg, Boutelle, Hannan, & Neumark-Sztainer, 2011; Wardle & Cooke, 2005), and poorer mental and physical health-related quality of life (Griffiths et al., 2016; Griffiths, Henshaw, McKay, & Dunn, 2017). For males, muscularity and masculinity are important considerations in the manifestation of body dissatisfaction. The ideal male body in Western countries is that of the “mesomorph”—a body characterized by overt muscularity, broad shoulders, and V-shaped torso that leads down to a narrow waist (Grogan & Richards, 2002). For many men, the attainment of this ideal is linked with masculinity, including, for example, superior virility and physical ability (Grogan & Richards, 2002; Pope, Phillips, & Olivardia, 2000). To fall short of achieving this muscular ideal may generate feelings of inadequacy, increased drive for muscularity, and body dissatisfaction (Grogan, 2016; Mills & D’Alfonso, 2007). In extreme cases, this can result in muscle dysmorphia—the pathological fixation that one’s body is not sufficiently lean and muscular (Leone, Sedory, & Gray, 2005; Olivardia, 2001; Pope et al., 2000). Of concern, recent evidence suggests that body dissatisfaction in the western male population is prevalent and increasing (Adams, Turner, & Bucks, 2005; Cohane & Pope, 2001; Frederick et al., 2007; McCabe & Ricciardelli, 2004; Watkins, Christie, & Chally, 2008). For example, Frederick et al. (2007) reported that among four large samples of U.S. undergraduate males, 51–71% were dissatisfied

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This study complies with APA ethical standards, and there are no conflicts of interest that may arise due to the development or publication of this research. Also, the article is not being considered elsewhere for publication nor has it been presented to the public before (nor will it be while it is under consideration by *Psychology of Men and Masculinity*). We declare we have no conflict of interest. We also declare that none of the NSM-M data have been previously presented.

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with their level of body fat and 90% wanted to increase their muscularity.

Research and clinical efforts to address body image-related disorders (e.g., eating disorders and body dysmorphic disorder) require validated instruments that quickly and accurately capture the experience of body dissatisfaction. Self-report questionnaires are the most common method of measuring body dissatisfaction. Typically, these questionnaires include a multitude of items that prompt individuals to self-assess cognitions, attitudes, and behaviors toward specific aspects of their body (Grogan, 2016). However, questionnaires are limited in terms of their typically long test administration time and reliance on literacy skills of the individual.

Figural rating scales offer an alternative to self-report questionnaires. Figural rating scales typically depict a series of body figures that vary in size and/or shape. Respondents indicate the two figures that best represent their current actual body and ideal body, respectively. Although figural rating scales may not be a direct measure of body dissatisfaction, actual-ideal discrepancies between the two chosen bodies have been conceptualized as a measure of body dissatisfaction in past literature, with greater discrepancies indicating greater levels of body dissatisfaction (Cohen et al., 2015; Cororve Fingeret, Gleaves, & Pearson, 2004; Gardner & Brown, 2010; Gleaves et al., 2000). In addition, discrepancy scores have been found to correlate with a number of related measures including drive for muscularity (Gillen & Markey, 2015; Hildebrandt, Langenbucher, & Schlundt, 2004; Novella, Gosselin, & Danowski, 2015) and body appreciation (Mutale, Dunn, Stiller, & Larkin, 2016; Novella et al., 2015; Swami, Salem, Furnham, & Tovée, 2008).

Figural rating scales afford several advantages over self-report questionnaires. First, figural rating scales are often quicker and simpler to administer because the respondent is only required to make two responses. This is particularly advantageous outside of research settings where time is a limited resource (e.g., the assessment of clients in a clinical setting). Second, figural rating scales have arguably greater ecological validity than self-report questionnaires insofar as participants can view and select the type of bodies they perceive as their actual and ideal, rather than relying on more ambiguous choice descriptors on questionnaires. Third, figural rating scales allow for a visual comparison of the relative influence that muscularity and body fat make to a given male's body concern. This is particularly helpful in aiding diagnosis and assessment of treatment progression of body image-related disorders.

One of the most prominent figural rating scales for men in the extant literature is the Somatomorphic Matrix (Pope et al., 2000). The Somatomorphic Matrix is a computerized body image test that can assess body image satisfaction with respect to two dimensions: *muscularity* and *body fat*. The male version of the test consists of a computerized library of 100 images of men, arranged in a 10 × 10 matrix, representing 10 degrees of body fat (*x*-axis) and 10 degrees of muscularity (*y*-axis). On the body fat axis, the figures begin at 4% body fat and increase in increments of 4% (Cafri & Thompson, 2004). On the muscularity axis, the images begin at a fat-free mass index (FFMI; Kouri, Pope, Katz, & Oliva, 1995) of 16.5 kg/m² (36.4 lbs/m²) and increase in increments of 1.5 kg/m² (3.3 lbs/m²). FFMI reflects body composition in terms of muscular bulk, and height, excluding mass due to body fat. Typically, FFMI ranges from 16 (35.3 lbs/m²; characteristic of a frail individual) to 25 (55.1 lbs/m²; an extremely muscular individual) in males. It has

been argued that, without the use exogenous substances, including steroids, an FFMI above 25 would be challenging to obtain (Pope et al., 2000).

The Somatomorphic Matrix has been used in various populations and in various countries, typically showing that college-aged men have a preference (and believe women have a preference) for an ideal body significantly more muscular than their current body (Frederick et al., 2007; Pope et al., 2000). A modified version of the Somatomorphic Matrix with a reduced amount of image variation (34 images instead of 100) was developed (Cafri & Thompson, 2004). Although the number of images were significantly reduced (depicting every third image of the original), the modified Somatomorphic Matrix permits equivalent response resolution (Cafri & Thompson, 2004). A limitation of the Somatomorphic Matrix (and modified version) lies in the quality and detail of the body images presented. The body images are hand-drawn and provide little detail in terms of muscle tone and shape, and are restricted in the number of muscle groups presented. In addition, not all of these hand-drawn body images are symmetrical. This has the potential to confound responses given that previous research has demonstrated the influence of asymmetry on ratings of attractiveness (Rhodes & Simmons, 2007; Tovée, Tasker, & Benson, 2000). This limits the ecological validity of the Somatomorphic Matrix, and thus may compromise the scale's ability to accurately measure individuals' actual and ideal body.

Studies examining body dissatisfaction and related disorders among sexual minority males are of particular interest. Previous research has demonstrated that sexual minority males have higher rates of body dissatisfaction (Frederick & Essayli, 2016) and eating disorders (Calzo, Blashill, Brown, & Argenal, 2017) compared with their heterosexual counterparts. These elevated rates of body dissatisfaction may stem from differences in body ideals between gay and straight populations, as there is some indication that sexual minority males tend to place more importance on muscularity as opposed to body fat, and possess body ideals that are more muscular compared with heterosexual men (Levesque & Vichesky, 2006; Tiggemann, Martins, & Kirkbride, 2007; Yelland & Tiggemann, 2003).

Pronounced muscular body ideals observed in sexual minority males might, in part, be explained by processes relating to masculinity. Pope et al. (2000) posited that sexual minority males might desire a more masculine physique compared with heterosexual men as a defensive reaction to mainstream society's stigmatization of them as "un-manly." As people tend to associate muscle with masculinity (Kimmel & Mahalik, 2004; McCreary & Sasse, 2000), sexual minority males may seek to compensate for the perception that they are inadequately masculine through acquiring high levels of muscularity. Alternatively, sexual minority males might aspire toward masculine bodies simply to appear more attractive to others. Yelland and Tiggemann (2003) found that sexual minority males are more likely to report that they exercise to enhance attractiveness, and that they perceived that their physical appearance was important to other people, compared with heterosexual men. In addition, in a qualitative study of 547 self-identifying gay men, pressure to appear masculine to be seen as desirable by other gay men was cited as a central factor associated with their own masculinity (Sánchez, Greenberg, Liu, & Vilain, 2009). This focus on masculinity through muscularity might behaviorally manifest as restrictive diets, greater frequency

of muscle building-focused exercise, and anabolic-androgenic steroid misuse. To this end, U.S.-based population studies have shown that sexual minority male adolescents are more likely to report lifetime history of steroid use compared with their heterosexual equivalents (Blashill, Calzo, Griffiths, & Murray, 2017; Blashill & Safren, 2014). Further, experimental studies have shown that men who engage in muscle-building eating and exercise behaviors are perceived as masculine by men and women (Griffiths, Mond, Murray, & Touyz, 2014). For these reasons, it is important to develop tools that are capable of accurately obtaining measures of body image and body dissatisfaction in sexual minority males, and are sensitive to detecting body image-related differences in male subgroups within this population (such as steroid users and obese groups).

The aim of our Study 1 was to provide validity evidence for the New Somatomorphic Matrix–Male (NSM-M) as a measure of actual–ideal male body discrepancy. To this end, we devised six hypotheses that, if supported by data, would provide evidence in favor of the validity of the NSM-M. We hypothesized as follows:

Hypothesis 1: Discrepancy scores on the muscularity dimension of the NSM-M would correlate positively with muscularity dissatisfaction scores on the self-report Male Body Attitudes Scale–Revised (MBAS-R; Ryan, Morrison, Roddy, & McCutcheon, 2011).

Hypothesis 2: Discrepancy scores on the body fat dimension of the NSM-M would correlate positively with body fat dissatisfaction scores on MBAS-R.

Hypothesis 3: The correlation of NSM-M muscularity discrepancy scores with MBAS-R muscularity scores would be significantly higher than the correlation of NSM-M muscularity discrepancy scores with MBAS-R body fat scores (and vice versa).

Hypothesis 4: NSM-M discrepancy scores would correlate positively with measures of psychopathology related to body dissatisfaction, including eating disorder symptoms and thoughts about using anabolic steroids, and negatively correlate with quality of life. In addition, these correlations would be maintained when controlling for MBAS-R body dissatisfaction scores.

Hypothesis 5: NSM-M muscle discrepancy scores would be greater for anabolic steroid users compared with nonusers.

Hypothesis 6: NSM-M body fat discrepancy scores would positively correlate with body mass index (BMI).

Support for each of our six hypotheses was intended to be construed as evidence in support of the validity of the NSM-M as a measure of actual–ideal discrepancy.

Study 2 aimed to assess the test–retest reliability of the NSM-M over a 7- to 14-day period. It was hypothesized that if the NSM-M was a reliable measure of actual–ideal discrepancy, then NSM-M scores from *Session 1* would return high correlations (above the recommended .80; Carmines, 1990) with NSM-M scores from *Session 2*.

Study 1

Method

Participants. The majority (99.1%) of participants identified their gender as male. Of the remaining participants, 0.4% identified as “gender-fluid,” “gender-queer,” “gender-neutral,” or “non-binary,” 0.3% as “transgender,” and 0.2% as “other.” Every participant reported that they identified as a minority sexual orientation, with 68.3% “exclusively gay/homosexual,” 21.3% “mostly gay/homosexual,” 8.6% “bisexual,” 1.1% “mostly straight/heterosexual,” 0.4% “other,” and 0.3% “pansexual.” A broad age range was observed, from 17 to 78 years ($M = 33.77$, $SD = 12.01$). The majority of participants resided in Australia (78.4%) and reported an “Australian” (50.4%) cultural/ethnic background. Other reported cultural/ethnic backgrounds included New Zealand (15.7%), North-West European (7.9%), South-East Asian (6.4%), Southern and Eastern European (3.5%), Southern and Central Asian (2.8%), and North-East Asian (2.6%). The remaining 10.7% of participants reported a variety of cultural backgrounds.

The total number of individuals that accessed the survey was $N = 3,768$. A total of 539 participants were excluded from the study, as they did not provide consent ($n = 397$) or data ($n = 142$). A further 452 were excluded due to high levels of missing data. In addition, five individuals were excluded due to obvious patterns of implausible data, and two individuals were excluded due to reporting female gender. The remaining 2,733 participants were included in the study. Participants’ age ranged from 18–75 ($M = 33.12$, $SD = 11.37$), and self-reported BMI ranged from 14.15–68.12 ($M = 25.90$, $SD = 5.58$; 3.0% Underweight, 51.5% healthy weight, 28.1% overweight, 11.1% Obese I, 4.2% Obese II, and 2.1% Obese III).

Procedure. Participants were recruited via an advertisement for “Gay Bodies—a study of body image and body change behaviors among gay men only,” which was placed on a popular geo-social networking app for sexual minority males in Australia and New Zealand. However, a large number of bisexual men were also recruited. Thus, we opted to use the more inclusive and appropriate term *sexual minority men*. Responding to the advertisement redirected potential participants to an information and consent page for an online survey. Consenting participants continued on to the survey, which took a median time of 11 min to complete. Ethical approval was received by the Behavioral and Social Sciences Human Ethics Subcommittee at the University of Melbourne (ethics ID: 1748752).

Measures.

Body mass index. Participants were asked to self-report their weight (kg) and height (in their preferred metric; 79.3% of participants responded in centimeters). Responses were converted to centimeters for the purposes of statistical analysis.

New Somatomorphic Matrix–Male. The figures presented in this matrix (Figure 1) were designed to emulate those present in the Modified Somatomorphic Matrix (Cafri & Thompson, 2004). These figures were rendered through DAZ Studio 4.9 (www.Daz3d.com), using the male 3D model Michael 6.0. To estimate body fat percentage and FFMI values for each figure, a pilot study was conducted in which randomly generated bodies (in term of body fat and muscularity; $n = 22$) were matched to the Cafri and

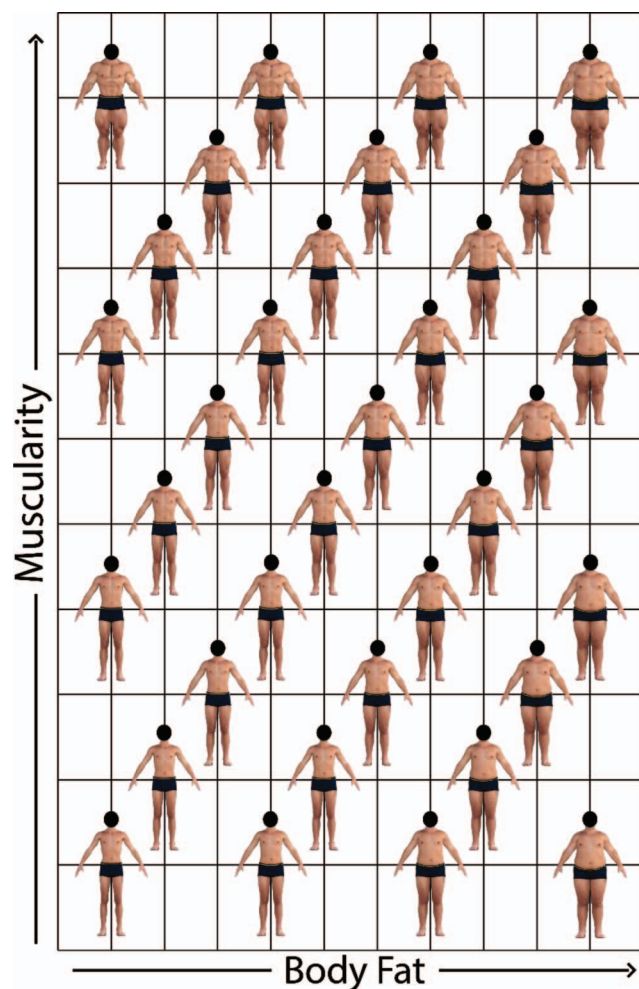


Figure 1. The New Somatomorphic Matrix–Male (NSM–M). The x -axis displays body fat percentage ranging from 4 to 40%, increasing in increments of 4%. The y -axis presents fat-free mass index ranging from 16.5 to 30 kg/m^2 (36.4–66.1 lbs/m^2), increasing in increments of 1.5 kg/m^2 . See the online supplemental materials for a higher resolution version of the NSM–M. See the online article for the color version of this figure.

Thompson's (2004) Modified Somatomorphic Matrix. This pilot study ensured that each rendered figure was perceptually equivalent to the Somatomorphic Matrix Modification (Cafri & Thompson, 2004). Body fat percentage and FFMI values for each figure were assigned accordingly. On the x -axis (body fat axis), body fat percentage ranges from 4–40%, increasing in increments of 4%. On the y -axis (muscularity axis), FFMI ranges from 16.5–30 kg/m^2 (36.4–66.1 lbs/m^2), increasing in increments of 1.5 kg/m^2 (3.3 lbs/m^2). Once presented with the NSM–M, participants were asked the following: (a) *Please touch the location on the grid that best corresponds to your actual body*, and (b) *Please touch the location on the grid that best corresponds to your ideal body (i.e., the body you would most like to have as your own)*.

Body dissatisfaction. The MBAS–R (Ryan et al., 2011) was used to measure body dissatisfaction. The MBAS–R is composed of 15 items, including three subscales measuring dissatisfaction with muscularity (seven items), body fat (five items), and height

(three items). Participants responded via a 5-point Likert scale ranging from *never* to *always*. In previous studies, the MBAS–R has demonstrated sound psychometric properties, with Cronbach's α s of .88 and .91 for the Muscularity and Body Fat subscales, respectively (Ryan et al., 2011). In the present study, Cronbach's α s were .88 and .92 for the Muscularity and Body Fat subscales, respectively.

Eating disorder symptoms. The Eating Disorders Examination Questionnaire Short (EDE–QS; Gideon et al., 2016) was used to measure eating disorder symptoms over the past 2 weeks. This included assessment of symptoms such as dietary restriction, binge eating, and overvaluation of weight/shape. Participants indicated frequency of symptoms on a 4-point Likert scale, ranging from 0 (0 days) to 3 (6–7 days). The EDE–QS was derived from the EDE–Q (Fairburn & Beglin, 1994) to provide a more concise scale and improve reliability. In a previous study the EDE–QS demonstrated construct validity, predictive validity, and test–retest reliability (intraclass correlation coefficient = .93, $p < .001$, time range: 2–14 days; Gideon et al., 2016). In the present study, the Cronbach's α was .88.

Thoughts about using anabolic steroids. To examine participants' thoughts about using anabolic steroids, they were asked the following: "How frequently do you think about using anabolic steroids?" with no time frame for steroid use specified. The structure of this question was based on recommendations made in previous research (Kanayama, Barry, Hudson, & Pope, 2006). First, the term *anabolic steroids* was used, as opposed to the more ambiguous term *steroids*; second, to reduce the chance of false-positive responses, participants were provided with three names of commonly used steroids and sympathomimetic amines in Australia and New Zealand (testosterone enanthate, dianabol, and clenbuterol; Griffiths et al., 2017). Response options ranged from 1 (*never*) through 6 (*always*).

Quality of life. Participants' quality of life was measured using the Brunnsviden Brief Quality of Life Scale (BBQ; Lindner et al., 2016). The 12-item BBQ assesses perceived level of satisfaction with six domains: leisure, creativity, learning, friends and friendship, view on one's life, and view on one's self. Participants respond via a 5-point scale ranging from 0 (*strongly disagree*) to 4 (*strongly agree*). Further, participants are also required to rank the importance of each domain (1–6) in determining their overall quality of life. For each domain, weighted scores are calculated by multiplying raw scale scores by the ranked level of importance ascribed to the relevant domain. A total quality of life score (ranging from 0 to 96) is calculated by summing all six scale scores. The BBQ has demonstrated a robust single-factor structure and acceptable internal consistency ($\alpha = .76$) and test–retest reliability (intraclass correlation coefficient = .82, $p < .001$, time range: 7 days $\pm$ 1 day) for the total score (Lindner et al., 2016). In the present study, the Cronbach's α was .81.

Statistical analysis. First, we conducted a Spearman's rank correlation to examine the association between NSM–M muscularity discrepancy scores and MBAS–R muscularity dissatisfaction scores. Second, another Spearman's rank correlation was conducted to examine the association between the NSM–M body fat discrepancy scores and the MBAS–R body fat dissatisfaction scores. This was taken as an appropriate measure of convergent validity, as NSM–M dissatisfaction scores and MBAS–R subscales

both provide measures of body fat and muscle dissatisfaction, respectively (Grogan, 2016; Ryan et al., 2011).

Third, to examine whether the NSM-M muscularity and body fat dimensions were independent of each other, we conducted Spearman's rank correlation analyses to examine the associations of (a) NSM-M muscularity discrepancy scores with MBAS-R body fat dissatisfaction scores, and (b) NSM-M body fat discrepancy scores with MBAS-R muscularity dissatisfaction scores, with the expectation that these scores would evidence weak or nonsignificant correlations. In addition, Fisher's transformations were conducted to convert these correlation coefficients to z scores, and two Steiger's z tests were conducted to examine (a) whether *NSM-M muscularity discrepancy–MBAS-R muscularity dissatisfaction* correlations were different from *NSM-M muscularity discrepancy–MBAS-R body fat dissatisfaction* correlations, and (b) whether *NSM-M body fat discrepancy–MBAS-R body fat dissatisfaction* correlations were different from *NSM-M body fat discrepancy–MBAS-R muscularity dissatisfaction* correlations. Previous studies have demonstrated that Steiger's z is an appropriate test for examining the difference between Spearman correlation coefficients (Myers & Sirois, 2006).

Fourth, to examine associations of NSM-M scores with body dissatisfaction, eating disorder symptoms, quality of life, and thoughts about steroid use, Spearman's correlational analyses using participants' NSM-M discrepancy scores, MBAS-R, EDE-QS, BBQ, and the thoughts about using anabolic steroids item were conducted. Significant correlations were taken as evidence of concurrent validity, given previous research showing associations of body dissatisfaction with eating disorder symptoms (Hospers & Jansen, 2005; Mayo & George, 2014; Olivardia et al., 2004), quality of life (Griffiths et al., 2016), and steroid use (Jenssen & Johannessen, 2015). In addition, partial correlations were used to examine whether NSM-M body fat discrepancy and NSM-M muscularity discrepancy scores were uniquely associated with the EDE-QS, BBQ, and thoughts about steroid use, after controlling for the MBAS-R Muscularity and Body Fat subscales. Partial correlations were only conducted in cases in which initial significant correlations between discrepancy and EDE-QS, BBQ, and thoughts about steroid use scores were present. This was taken as an examination of incremental validity evidence (i.e., whether the NSM-M was significantly associated with eating disorder symptoms, quality of life, and thoughts about steroid use beyond that of the MBAS-R).

Fifth, a series of moderation analyses were conducted to examine whether cultural/ethnic background (Australian/New Zealander vs. other) moderated associations between NSM-M discrepancy scores and MBAS-R, EDE-QS, BBQ, and thoughts about anabolic steroid scores. Participants who identified Australia or New Zealand as their main cultural/ethnic group were coded as Australian/New Zealander. Participants who identified as belonging to a cultural/ethnic group that is predominantly non-Caucasian (e.g., South-East Asian) were coded as other. Sixth, a series of six Mann–Whitney U tests were conducted to examine whether scores on the NSM-M would discriminate between steroid users and nonusers. Seventh, two Spearman's rank correlations were conducted to examine associations between NSM-M body fat discrepancy scores and BMI, and NSM-M muscularity dissatisfaction scores and BMI, respectively.

Results

Descriptives for participants' NSM-M "actual" body, NSM-M "ideal" body, and NSM-M discrepancy scores are shown in Table 1. A schematic of "actual" and "ideal" body responses are shown in Figure 2.

NSM-M and body dissatisfaction. Results showed that NSM-M discrepancy scores for body fat were significantly positively correlated with MBAS-R body fat dissatisfaction, and NSM-M discrepancy scores for muscularity were significantly positively correlated with MBAS-R muscularity dissatisfaction. Further, NSM-M discrepancy scores for body fat were significantly negatively correlated with MBAS-R muscularity dissatisfaction, and NSM-M muscularity discrepancy scores were significantly negatively correlated with MBAS-R body fat dissatisfaction (correlations are shown in Table 2). However, *NSM-M muscularity discrepancy–MBAS-R muscularity dissatisfaction* yielded a significantly stronger correlation than *NSM-M muscularity discrepancy–MBAS-R body fat dissatisfaction*, $z = 17.638, p < .001$. Further, *NSM-M body fat discrepancy–MBAS-R body fat dissatisfaction* yielded a significantly stronger correlation than *NSM-M body fat discrepancy–MBAS-R muscularity dissatisfaction*, $z = 31.513, p < .001$. Finally, NSM-M body fat discrepancy and NSM-M muscularity discrepancy were significantly positively correlated, $r = .097, p < .001$.

NSM-M and eating disorder symptoms, quality of life, and thoughts about steroid use. Results showed that participants' NSM-M body fat discrepancy scores were positively correlated with eating disorder symptoms, and negatively correlated with quality of life and frequency of thoughts about steroid use. Participants' NSM-M muscularity discrepancy scores were positively correlated with frequency of thoughts about steroid use, and negatively correlated with body fat dissatisfaction, eating disorder symptoms, and quality of life. Correlations are shown in Table 2.

In addition, partial correlations revealed that when controlling for MBAS-R body fat dissatisfaction, NSM-M body fat discrepancy scores were significantly correlated with quality of life, $r = -.043, p < .05$, but not with EDE-QS scores, $r = .015, p = .434$. When controlling for MBAS-R muscularity dissatisfaction, NSM-M muscularity discrepancy scores showed no significant

Table 1
Means, Standard Deviation, Median, and Range of New Somatomorphic Matrix–Male "Actual," "Ideal," and Actual–Ideal Discrepancy Ratings

Measures	M ($n = 2,733$)	SD	Median	Range
Body Fat				
Actual	20.52	11.09	20.52	0.57–44
Ideal	11.79	7.13	11.22	0.5–40.47
Discrepancy	8.72	10.12	8.10	0.0–40.62
Muscularity				
Actual	20.88	3.10	20.82	15.37–31.40
Ideal	24.80	3.04	25.38	15.11–31.45
Discrepancy	3.93	3.16	4.20	0–14.02

Note. Rows relating to body fat are reported in body fat percentage. Rows relating to muscularity are reported in fat-free mass index (kg/m^2). Actual = selected perceived actual body; Ideal = selected perceived ideal body; Discrepancy = difference score between selected actual and ideal body.

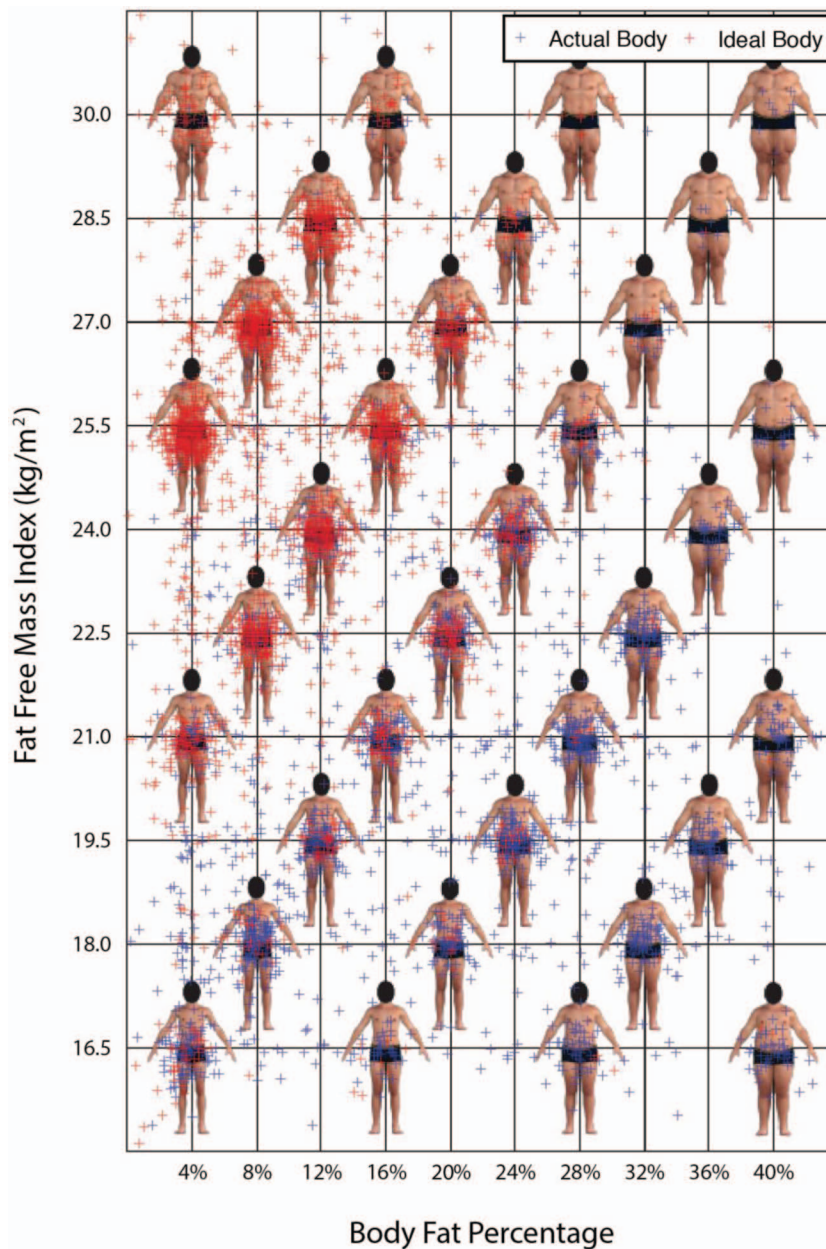


Figure 2. A plot of New Somatomorphic Matrix–Male “actual” (blue) and “ideal” (red) body scores. The x-axis displays body fat percentage ranging from 4 to 40%, increasing in increments of 4%. The y-axis presents fat-free mass index ranging from 16.5 to 30 kg/m² (36.4–66.1 lbs/m²), increasing in increments of 1.5 kg/m². See the online article for the color version of this figure.

correlations with quality of life, $r = -.026$, $p = .165$, or thoughts about steroid use, $r = .020$, $p = .298$.

NSM-M and Australian/New Zealander versus other participants. A series of moderation analyses were conducted to assess whether ethnicity (divided into Australian/New Zealander vs. other groups) moderated associations between NSM-M discrepancy scores and MBAS-R, EDE-QS, BBQ, and thoughts about anabolic steroid scores. Results showed that ethnicity significantly moderated the relationship between NSM-M body fat discrepancy and MBAS-R body fat dissatisfaction scores, $t(2185) = -2.624$,

$p = .009$. However, simple slopes analyses revealed that there was a significant association between NSM-M body fat discrepancy and MBAS-R body fat dissatisfaction for both Caucasian, $t(2185) = 36.47$, $p < .001$, and non-Caucasian, $t(2185) = 16.23$, $p < .001$, participants, respectively.

Results revealed no moderation effect for associations between NSM-M body fat discrepancy scores and EDE-QS ($p = .409$), BBQ ($p = .410$), and thoughts about anabolic steroid scores ($p = .175$). In addition, results revealed no moderation effect for associations between NSM-M muscularity discrepancy scores and

Table 2

Spearman's Rank Correlational Analysis of New Somatomorphic Matrix–Male (NSM-M) Actual–Ideal Discrepancy Scores, and Measures of Body Dissatisfaction, Eating Disorder Symptoms, Quality of Life, and Thoughts About Steroid Use

Measures	NSM-M discrepancy scores	
	Body fat (x-axis)	Muscularity (y-axis)
MBAS-R		
Muscularity	–.038*	.444***
Body Fat	.661***	–.052*
EDE-QS	.478***	–.003
BBQ	–.109***	–.099***
Thoughts about AAS	–.010	.123***

Note. MBAS-R = Male Body Attitudes Scale–Revised; EDE-QS = Eating Disorder Examination Questionnaire Short; BBQ = Brunnsvikien Brief Quality of Life Scale; AAS = anabolic–androgenic steroids; discrepancy scores = difference score between selected “actual” and “ideal” body. * $p < .05$. ** $p < .01$. *** $p < .001$.

MBAS-R body fat dissatisfaction ($p = .691$), EDE-QS ($p = .559$), BBQ ($p = .753$), and thoughts about anabolic steroid scores ($p = .271$).

NSM-M and steroid users versus nonusers. A series of Mann–Whitney U tests were conducted to assess whether NSM-M scores would distinguish anabolic steroid users from nonusers. Results (displayed in Table 3 revealed that steroid users’ perceived “actual” body was significantly more muscular, compared with nonusers’ perceived “actual” body. Further, steroid users reported significantly higher “ideal” bodies compared with nonusers.

NSM-M and BMI. Spearman’s rank correlational analysis was used to compare the ability of the NSM-M to detect body dissatisfaction across BMI categories. NSM-M discrepancy scores were correlated with BMI. Results showed that NSM-M body fat discrepancy scores positively correlated with BMI, $r = .560$, $p < .001$, and NSM-M muscularity discrepancy scores negatively correlated with BMI, $r = -.228$, $p < .001$. Correlations are represented in Figure 3.

Discussion

Study 1 aimed to develop and validate the NSM-M as a measure of male body dissatisfaction. To this end, NSM-M scores were compared with measures of body dissatisfaction, eating disorder symptoms, quality of life, and thoughts about steroid use. Additionally, the validity of the NSM-M depending on participants’ ethnicity, BMI, and steroid use was also considered.

NSM-M and body dissatisfaction. As hypothesized, NSM-M muscularity discrepancy scores were positively correlated with muscularity dissatisfaction scores on the MBAS-R (Ryan et al., 2011), and NSM-M body fat discrepancy scores were positively correlated with body fat dissatisfaction scores on MBAS-R. Consistent with our third hypothesis, NSM-M muscularity discrepancy scores were uncorrelated with MBAS-R body fat dissatisfaction scores and NSM-M body fat discrepancy were only very weakly associated with muscularity dissatisfaction scores. In addition, correlations between congruent NSM-M discrepancy scores and MBAS-R scores were significantly higher than incongruent NSM-M dissatisfaction and MBAS-R scores. Essentially, these

results showed that the NSM-M muscularity discrepancy score was sensitive in specifically detecting muscularity dissatisfaction, and the NSM-M body fat discrepancy score was sensitive in specifically detecting body fat dissatisfaction.

NSM-M and eating disorder symptoms, quality of life, and thoughts about steroid use. Results showed partial support for Hypothesis 4: NSM-M body fat discrepancy scores were significantly positively correlated with eating disorder symptoms. This likely reflects the association between body dissatisfaction and eating disorder symptoms (Hospers & Jansen, 2005; Mayo & George, 2014; Olivardia et al., 2004) and thus provides concurrent validity evidence for NSM-M body fat discrepancy measure. Conversely, the NSM-M muscularity discrepancy scores did not significantly correlate with eating disorder symptoms. These results could reflect the EDE-QS’s lack of sensitivity for detecting eating disorder symptoms in males. The EDE-QS is derived from the EDE-Q (Fairburn & Beglin, 1994), a measure which typically returns lower scores for males compared with females (Lavender, De Young, & Anderson, 2010; Reas, Øverås, & Rø, 2012). These lower scores could be due to a lack of sensitivity for detecting muscularity-related eating behaviors and body shape/weight attitudes. This would be exacerbated in the current sample, as sexual minority males tend to place a higher importance on muscularity compared with body fat (Levesque & Vichesky, 2006; Tiggemann et al., 2007).

As predicted, as the discrepancy between NSM-M “actual” and “ideal” bodies increased, participants’ scores for quality of life decreased. This is in line with previous studies that show analogous associations between body dissatisfaction and quality of life (Griffiths et al., 2016. Mond et al., 2013), thus providing additional concurrent validity evidence for the NSM-M.

Results also showed that participants who were further away from their muscular ideal (as recorded on the NSM-M) reported more thoughts about using steroids. This is comparable with previous studies who have demonstrated a link between steroids use and drive for muscularity (Peters, Copeland, & Dillon, 1999; Wright, Grogan, & Hunter, 2000), and body dissatisfaction (Blouin & Goldfield, 1995; Kanayama et al., 2006).

The NSM-M also showed some indication of incremental validity evidence, with a significant negative correlation between the

Table 3

Means and Mann–Whitney U Statistics of Steroid Users vs. Nonsteroid Users

Measures	Nonsteroid users ($n = 2,144$)	Steroid users ($n = 135$)	U
Body Fat			
Actual	20.56	20.12	165,663.50
Ideal	11.76	11.14	162,061.50
Discrepancy	8.75	8.97	173,162.50
Muscularity			
Actual	20.88	22.44	137,374.50***
Ideal	24.80	26.62	162,061.50***
Discrepancy	3.93	4.18	163,933.50

Note. Rows relating to body fat are reported in body fat percentage. Rows relating to muscularity are reported in fat-free mass index units (kg/m^2). Actual = selected perceived actual body; Ideal = selected perceived ideal body; Discrepancy = difference score between selected actual and ideal body.

*** $p < .001$.

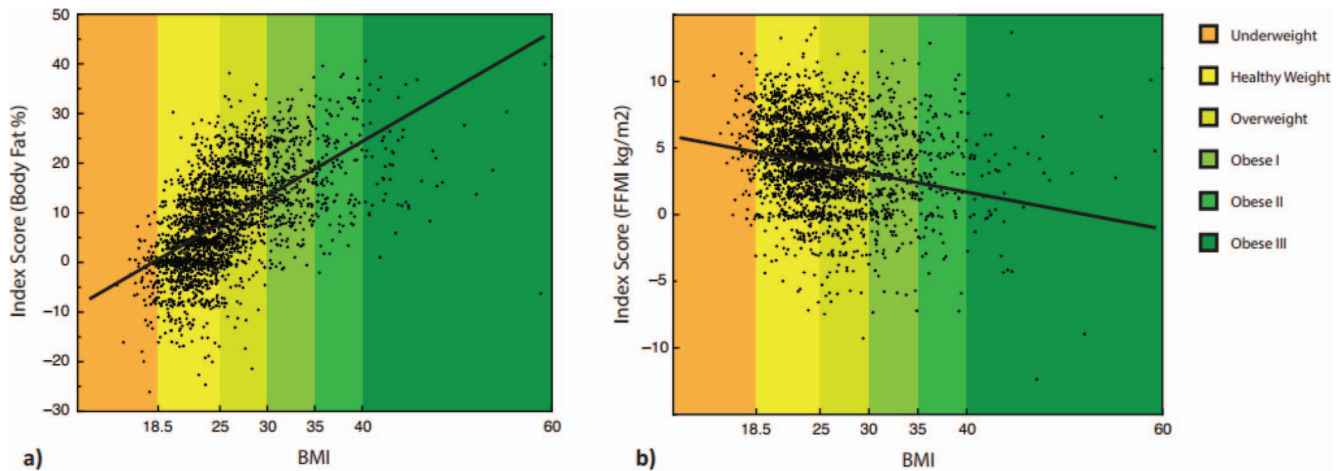


Figure 3. Correlations of New Somatomorphic Matrix–Male body dissatisfaction scores and BMI. (a) The x-axis displays BMI, and y-axis displays body fat percentage. (b) The x-axis displays BMI, and y-axis displays fat-free mass index (kg/m^2). The least square line of best fit is also displayed for each correlation. See the online article for the color version of this figure.

NSM-M body fat discrepancy score and the BBQ after controlling for scores on the MBAS-R. This suggests that the NSM-M holds unique associations with quality of life and has predictive ability for quality of life beyond that of the MBAS-R. However, no further incremental validity evidence was found.

NSM-M and Australian/New Zealander participants versus other participants. Given that the NSM-M uses images of males that may appear to have a Caucasian skin tone, it was important to consider the validity of this tool depending on ethnicity. Results revealed that ethnicity did not significantly moderate the relationship between the NSM-M discrepancy scores and measures of body dissatisfaction, eating disorder symptoms, quality of life, or thoughts about steroid use. The only significant moderation that was found was between NSM-M body fat discrepancy scores and body fat dissatisfaction scores, suggesting that this relationship was stronger for Australian/New Zealander participants. However, post hoc analysis revealed that strong significant relationships existed in both groups. These results suggest that despite the skin tone of the included images, the NSM-M was robust irrespective of whether or not participants identified as Australian or New Zealander.

NSM-M and steroid users versus nonusers. The fifth hypothesis was supported, as the NSM-M detected differences between steroid users and nonusers in perceived “actual” and “ideal” bodies. Participants who reported that they used steroids perceived their bodies as more muscular compared with nonusers. Further, steroid users held higher muscular ideals compared with nonusers. Of note, steroid users’ ideal muscularity exceeds $25 \text{ kg}/\text{m}^2$ ($55.1 \text{ lbs}/\text{m}^2$)—a level of muscularity that would be challenging to obtain without engaging in the use of exogenous substances, including steroids (Pope et al., 2000). Given that the NSM-M was able to identify elevated “actual” and “ideal” bodies in steroid users, it is possible that participants in this study were equally satisfied (or dissatisfied) with their level of muscularity irrespective of anabolic steroid use.

NSM-M and BMI. Results showed significant correlations between BMI and both NSM-M discrepancy scores, thus support-

ing the final hypothesis. Participants with higher BMIs (those falling within the obese range—BMI > 30) had a greater discrepancy between their “actual” and “ideal” body fat, compared with healthy weight (BMI = 18.5–24.9) and underweight participants (BMI < 18.5). This result likely reflects elevated levels of body dissatisfaction in obese compared with nonobese groups, as demonstrated in previous studies (Gavin, Simon, & Ludman, 2010; Mond et al., 2011; Wardle & Cooke, 2005). Further, results showed that those with lower a BMI tended to have greater discrepancy between their “actual” and “ideal” muscularity compared with those with a higher BMI. Given that it does not discriminate between fat and muscular mass, BMI can also be taken as a moderate measure of muscularity in males. Therefore, the association between BMI and NSM-M muscularity discrepancy scores likely reflects that less-muscular participants have a desire for a greater magnitude of muscular gain compared with more muscular (and thus higher BMI) participants. Of note, BMI displayed a stronger correlation with the NSM-M body fat discrepancy compared with the NSM-M muscularity discrepancy. This suggests that the NSM-M is more sensitive in detecting body fat compared with muscularity-related dissatisfaction across weight categories. These results show that the NSM-M is sensitive for detecting the varying levels of body dissatisfaction across the various BMI categories.

Study 2

Study 2 aimed to assess the test–retest reliability of the NSM-M over a 7- to 14-day period. If the NSM-M was a reliable measure of actual–ideal discrepancy, then NSM-M scores from Session 1 should return high correlations with NSM-M scores from Session 2.

Method

Participants. Thirty-eight Australian males participated in the study. From the sample, 57% of participants identified as Caucasian, 16% of participants identified as North or South-East Asian,

8% of participants identified as Southern or Central Asian, 8% of participants identified as African or Middle Eastern, and 11% of participants identified as other. Three participants self-identified as “homosexual,” whereas the remaining 30 participants self-identified as “heterosexual.” To compensate for their time, participants received a \$10AUD gift voucher.

Three participants were excluded from Study 2, as they failed to participate in both testing sessions. A further two participants were excluded as the time interval between completion of *Session 1* and *Session 2* exceeded 14 days. The remaining 33 participants were included in this study (age range = 18–35, $M = 23.52$, $SD = 6.66$; BMI range = 17–40, $M = 25.01$, $SD = 5.43$).

Procedure. Participants were recruited through advertising on noticeboards at an Australian University Campus. Participants e-mailed researchers directly if interested in participating in the study, and then were sent a link via e-mail to an online information page and consent form. Consenting participants continued on to *Session 1* of the study. Seven days after *Session 1*, participants were sent a second direct link via e-mail to *Session 2* of the study. Participants were asked to complete *Session 2* within 7 days of receiving the *Session 2* link. Each session took ~3 min to complete. Ethical approval to conduct Study 2 was provided by the Western Sydney University Human Research Ethics Committee (ethics ID: H11778).

Measures. In each session, participants completed a four-item demographic questionnaire enquiring about weight, height, ethnicity, and sexual identity. Additionally, the “actual” and “ideal” body ratings using the NSM-M were completed in each session (see Study 1).

Statistical analysis. To examine test–retest reliability of the NSM-M, intraclass correlations were conducted to assess associations between *Session 1* and *Session 2* scores. Intraclass correlational values above .80 were considered reliable, as recommended by Carmines (1990).

Results

Table 4 shows intraclass correlations between NSM-M “actual,” “ideal,” and discrepancy scores (body fat and muscularity) between testing *Session 1* and *Session 2*. “Actual,” “ideal,” and

discrepancy scores from *Session 1* were all significantly positively correlated with their equivalent scores from *Session 2*.

Discussion

Study 2 aimed to assess the test–retest reliability of the NSM-M over a 7- to 14-day period. Results suggested that NSM-M body fat “actual,” “ideal,” and NSM-M discrepancy scores all presented excellent test–retest reliability for the current sample. All three NSM-M body fat measures were found to be above the test–retest reliability standard of .80 (Carmines, 1990). Further, two out of three NSM-M muscularity scores exceeded the .80 standard, with only “actual” scores falling marginally below. These results suggest that, with the exception of the muscularity “actual” score, the NSM-M presents sound test–retest reliability. This shows a marked improvement on reported reliability scores for the Somatomorphic Matrix (which ranged from $r = .34$ to $.79$; Cafri, Roehrig, & Thompson, 2004).

General Discussion

Overall, the NSM-M displayed sound psychometric properties as a measure of actual–ideal body discrepancy. The NSM-M displayed excellent convergent validity evidence, as NSM-M discrepancy scores were highly correlated with appropriate measures of body dissatisfaction (specific to body fat and muscularity). The NSM-M displayed convergent validity evidence through correlations between NSM-M discrepancy scores and eating disorder symptoms, quality of life, and thoughts about steroid use. In addition, the NSM-M demonstrated some incremental validity, showing predictive ability for quality of life beyond that of the MBAS-R. The NSM-M also demonstrated the ability to detect group differences in body image between steroid users and non-users, and at varying levels of BMI, respectively. In addition, the NSM-M proved to be a reliable tool. Five out of six measures returned *Session 1*–*Session 2* correlations above the .80 standard (Carmines, 1990).

As well as being a sensitive measure of actual–ideal body discrepancy, the NSM-M also demonstrates advantages over questionnaires including being quicker and simpler to administer. The NSM-M was also designed to have compatible display for smartphones, as well as laptop/desktop computers, and also features zoom functionality for both Android and iOS to allow respondents to make more fine-grained selections. Further, the NSM-M presents greater ecological validity compared with past versions of the Somatomorphic Matrix through use of improved body stimuli. These new set of rendered body stimuli are far more anatomically representative compared with those presented in Cafri and Thompson’s (2004) Modified Somatomorphic Matrix, and thus should enhance the validity of the tool for capturing body image and body dissatisfaction. The NSM-M also presents symmetrical stimuli, thus avoiding any biases caused by the influence of asymmetry (Rhodes & Simmons, 2007; Tovée et al., 2000).

Limitations of the study are noted. First, the bodies depicted in the NSM-M are of Caucasian skin pigmentation. This potentially limits the ability of individuals from non-Caucasian populations to identify with the presented bodies, and thus, may impair the ecological validity of the NSM-M. Although our study provided evidence that participants’ ethnicity did not meaningfully influ-

Table 4
Means, F Values, P Values, Intraclass Correlations, and Confidence Intervals for Session 1 and Session 2 New Somatomorphic Matrix–Male (NSM-M) Scores

Measures	Session 1 <i>M</i>	Session 2 <i>M</i>	ICC	95% CI
NSM-M Body Fat				
Actual	18.50	18.79	.93***	[.85, .96]
Ideal	10.92	12.47	.91***	[.81, .95]
Discrepancy	4.58	3.12	.86***	[.71, .93]
NSM-M Muscularity				
Actual	20.21	20.52	.78***	[.55, .89]
Ideal	23.69	24.51	.95***	[.90, .98]
Discrepancy	3.48	3.99	.84***	[.67, .92]

Note. ICC = intraclass correlations; CI = confidence interval; Actual = selected perceived “actual” body; Ideal = selected perceived “ideal” body; Discrepancy = difference score between selected “actual” and “ideal” body.
*** $p < .001$.

ence results, a better approach would have been to assess whether race moderated NSM-M responses (race was not recorded for Study 1). Future studies may wish to examine whether the use of gray-scale (i.e., black-and-white) version of the NSM-M, which might, at least to some extent, ameliorate the potential issue of skin color to some extent. An additional, more sophisticated, approach, may be to develop complementary versions of the NSM-M for individuals from specific ethnic backgrounds. Second, Study 1 focused on sexual minority men, whereas Study 2 focused on heterosexual men. Although it is highly unlikely that the relationships between test-retest reliability findings in the present study would be notably different between the two populations, it would be preferable to run both Study 1 and Study 2 in dual samples of heterosexual and sexual minority men. Future research should aim to examine the validity evidence of the NSM-M using a heterosexual sample. A third potential limitation is the use of a geo-social networking app to recruit participants for Study 1. In a previous study, Strubel and Petrie (2017) demonstrated that dating app users tend to report higher levels of body dissatisfaction, appearance comparisons, and body shame compared with nonusers. Further, sexual minority men tend to visually privilege the presentation of their own masculinity in their social networking app profile (Miller, 2015). Therefore, our recruitment method may have produced a sample with inflated body image concern. A fourth limitation was the use of BMI to examine body composition. When considering self-report BMI, it is difficult to determine whether BMI value is driven by fat or muscle mass, or a combination of both. This could have complications for analyses relating to Hypothesis 6. Future studies should aim to provide validity evidence for the NSM-M through examining associations between the NSM-M and more appropriate measures of male body composition (e.g., body fat percentage and FFMI). Future use of the NSM-M by other researchers, and by clinicians involved in the treatment of individuals with body image and related disorders, may yield additional evidence for the validity of the NSM-M.

References

- Adams, G., Turner, H., & Bucks, R. (2005). The experience of body dissatisfaction in men. *Body Image*, 2, 271–283. <http://dx.doi.org/10.1016/j.bodyim.2005.05.004>
- Blashill, A. J., Calzo, J. P., Griffiths, S., & Murray, S. B. (2017). Anabolic steroid misuse among U.S. adolescent boys: Disparities by sexual orientation and race/ethnicity. *American Journal of Public Health*, 107, 319–321. <http://dx.doi.org/10.2105/AJPH.2016.303566>
- Blashill, A. J., & Safren, S. A. (2014). Sexual orientation and anabolic-androgenic steroids in U.S. adolescent boys. *Pediatrics*, 133, 469–475. <http://dx.doi.org/10.1542/peds.2013-2768>
- Blouin, A. G., & Goldfield, G. S. (1995). Body image and steroid use in male bodybuilders. *International Journal of Eating Disorders*, 18, 159–165. [http://dx.doi.org/10.1002/1098-108X\(199509\)18:2<159::AID-EAT2260180208>3.0.CO;2-3](http://dx.doi.org/10.1002/1098-108X(199509)18:2<159::AID-EAT2260180208>3.0.CO;2-3)
- Cafri, G., Roehrig, M., & Thompson, J. K. (2004). Reliability assessment of the Somatomorphic Matrix. *International Journal of Eating Disorders*, 35, 597–600. <http://dx.doi.org/10.1002/eat.10270>
- Cafri, G., & Thompson, J. K. (2004). Measuring male body image: A review of the current methodology. *Psychology of Men and Masculinity*, 5, 18–29. <http://dx.doi.org/10.1037/1524-9220.5.1.18>
- Calzo, J. P., Blashill, A. J., Brown, T. A., & Argenal, R. L. (2017). Eating disorders and disordered weight and shape control behaviors in sexual minority populations. *Current Psychiatry Reports*, 19, 49. <http://dx.doi.org/10.1007/s11920-017-0801-y>
- Carmines, E. G. (1990). The statistical analysis of overidentified linear recursive models. *Quality and Quantity: International Journal of Methodology*, 24, 65–85. <http://dx.doi.org/10.1007/BF00221385>
- Cohane, G. H., & Pope, H. G., Jr. (2001). Body image in boys: A review of the literature. *International Journal of Eating Disorders*, 29, 373–379. <http://dx.doi.org/10.1002/eat.1033>
- Cohen, E., Bernard, J. Y., Ponty, A., Ndao, A., Amougou, N., Saïd-Mohamed, R., & Pasquet, P. (2015). Development and validation of the body size scale for assessing body weight perception in African populations. *PLoS ONE*, 10, e0138983. <http://dx.doi.org/10.1371/journal.pone.0138983>
- Cororve Fingeret, M., Gleaves, D. H., & Pearson, C. A. (2004). On the methodology of body image assessment: The use of figural rating scales to evaluate body dissatisfaction and the ideal body standards of women. *Body Image*, 1, 207–212. <http://dx.doi.org/10.1016/j.bodyim.2004.01.003>
- Fairburn, C. G., & Beglin, S. J. (1994). Assessment of eating disorders: Interview or self-report questionnaire? *International Journal of Eating Disorders*, 16, 363–370.
- Frederick, D. A., Buchanan, G. M., Sadehgi-Azar, L., Peplau, L. A., Haselton, M. G., Berezovskaya, A., & Lipinski, R. E. (2007). Desiring the muscular ideal: Men's body satisfaction in the United States, Ukraine, and Ghana. *Psychology of Men and Masculinity*, 8, 103–117. <http://dx.doi.org/10.1037/1524-9220.8.2.103>
- Frederick, D. A., & Essayli, J. H. (2016). Male body image: The roles of sexual orientation and body mass index across five national U.S. studies. *Psychology of Men and Masculinity*, 17, 336–351. <http://dx.doi.org/10.1037/men0000031>
- Gardner, R. M., & Brown, D. L. (2010). Body image assessment: A review of figural drawing scales. *Personality and Individual Differences*, 48, 107–111. <http://dx.doi.org/10.1016/j.paid.2009.08.017>
- Gavin, A. R., Simon, G. E., & Ludman, E. J. (2010). The association between obesity, depression, and educational attainment in women: The mediating role of body image dissatisfaction. *Journal of Psychosomatic Research*, 69, 573–581. <http://dx.doi.org/10.1016/j.jpsychores.2010.05.001>
- Gideon, N., Hawkes, N., Mond, J., Saunders, R., Tchanturia, K., & Serpell, L. (2016). Development and psychometric validation of the EDE-QS, a 12 item short form of the eating disorder examination questionnaire (EDE-Q). *PLoS ONE*, 11, e0152744. <http://dx.doi.org/10.1371/journal.pone.0152744>
- Gillen, M., & Markey, C. N. (2015). Development and validation of the muscle pictorial measure. *Archives of Assessment Psychology*, 5, 11–22.
- Gleaves, D. H., Cepeda-Benito, A., Williams, T. L., Cororve, M. B., Fernandez, M. C., & Vila, J. (2000). Body image preferences of self and others: A comparison of Spanish and American male and female college students. *Eating Disorders: The Journal of Treatment and Prevention*, 8, 269–282. <http://dx.doi.org/10.1080/10640260008251236>
- Griffiths, S., Hay, P., Mitchison, D., Mond, J. M., McLean, S. A., Rodgers, B., . . . Paxton, S. J. (2016). Sex differences in the relationships between body dissatisfaction, quality of life and psychological distress. *Australian and New Zealand Journal of Public Health*, 40, 518–522. <http://dx.doi.org/10.1111/1753-6405.12538>
- Griffiths, S., Henshaw, R., McKay, F. H., & Dunn, M. (2017). Post-cycle therapy for performance and image enhancing drug users: A qualitative investigation. *Performance Enhancement and Health*, 5, 103–107. <http://dx.doi.org/10.1016/j.peh.2016.11.002>
- Griffiths, S., Mond, J. M., Murray, S. B., & Touyz, S. (2014). Young peoples' stigmatizing attitudes and beliefs about anorexia nervosa and muscle dysmorphia. *International Journal of Eating Disorders*, 47, 189–195. <http://dx.doi.org/10.1002/eat.22220>

- Grogan, S. (2016). *Body image: Understanding body dissatisfaction in men, women and children*. Abingdon, United Kingdom: Taylor & Francis.
- Grogan, S., & Richards, H. (2002). Body image: Focus groups with boys and men. *Men and Masculinities*, 4, 219–232. <http://dx.doi.org/10.1177/1097184X02004003001>
- Hildebrandt, T., Langenbucher, J., & Schlundt, D. G. (2004). Muscularity concerns among men: Development of attitudinal and perceptual measures. *Body Image*, 1, 169–181. <http://dx.doi.org/10.1016/j.bodyim.2004.01.001>
- Hospers, H. J., & Jansen, A. (2005). Why homosexuality is a risk factor for eating disorders in males. *Journal of Social and Clinical Psychology*, 24, 1188–1201. <http://dx.doi.org/10.1521/jscp.2005.24.8.1188>
- Jenssen, I. H., & Johannessen, K. B. (2015). Aggression and body image concerns among anabolic androgenic steroid users, contemplators, and controls in Norway. *Body Image*, 12, 6–13. <http://dx.doi.org/10.1016/j.bodyim.2014.08.009>
- Kanayama, G., Barry, S., Hudson, J. I., & Pope, H. G., Jr. (2006). Body image and attitudes toward male roles in anabolic-androgenic steroid users. *The American Journal of Psychiatry*, 163, 697–703. <http://dx.doi.org/10.1176/ajp.2006.163.4.697>
- Kimmel, S. B., & Mahalik, J. R. (2004). Measuring masculine body ideal distress: Development of a measure. *International Journal of Men's Health*, 3, 1–10. <http://dx.doi.org/10.3149/jmh.0301.1>
- Kouri, E. M., Pope, H. G., Jr., Katz, D. L., & Oliva, P. (1995). Fat-free mass index in users and nonusers of anabolic-androgenic steroids. *Clinical Journal of Sport Medicine: Official Journal of the Canadian Academy of Sport Medicine*, 5, 223–228. <http://dx.doi.org/10.1097/00042752-199510000-00003>
- Lavender, J. M., De Young, K. P., & Anderson, D. A. (2010). Eating Disorder Examination Questionnaire (EDE-Q): Norms for undergraduate men. *Eating Behaviors*, 11, 119–121. <http://dx.doi.org/10.1016/j.eatbeh.2009.09.005>
- Leone, J. E., Sedory, E. J., & Gray, K. A. (2005). Recognition and treatment of muscle dysmorphia and related body image disorders. *Journal of Athletic Training*, 40, 352–359.
- Levesque, M. J., & Vichesky, D. R. (2006). Raising the bar on the body beautiful: An analysis of the body image concerns of homosexual men. *Body Image*, 3, 45–55. <http://dx.doi.org/10.1016/j.bodyim.2005.10.007>
- Lindner, P., Frykheden, O., Forsström, D., Andersson, E., Ljótsson, B., Hedman, E., . . . Carlbring, P. (2016). The Brunnsvikens Brief Quality of Life Scale (BBQ): Development and psychometric evaluation. *Cognitive Behaviour Therapy*, 45, 182–195. <http://dx.doi.org/10.1080/16506073.2016.1143526>
- Mayo, C., & George, V. (2014). Eating disorder risk and body dissatisfaction based on muscularity and body fat in male university students. *Journal of American College Health*, 62, 407–415. <http://dx.doi.org/10.1080/07448481.2014.917649>
- McCabe, M. P., & Ricciardelli, L. A. (2004). Body image dissatisfaction among males across the lifespan: A review of past literature. *Journal of Psychosomatic Research*, 56, 675–685. [http://dx.doi.org/10.1016/S0022-3999\(03\)00129-6](http://dx.doi.org/10.1016/S0022-3999(03)00129-6)
- McCreary, D. R., & Sasse, D. K. (2000). An exploration of the drive for muscularity in adolescent boys and girls. *Journal of American College Health*, 48, 297–304. <http://dx.doi.org/10.1080/07448480009596271>
- Miller, B. (2015). “Dude, where’s your face?” Self-presentation, self-description, and partner preferences on a social networking application for men who have sex with men: A content analysis. *Sexuality and Culture*, 19, 637–658. <http://dx.doi.org/10.1007/s12119-015-9283-4>
- Mills, J. S., & D’Alfonso, S. R. (2007). Competition and male body image: Increased drive for muscularity following failure to a female. *Journal of Social and Clinical Psychology*, 26, 505–518. <http://dx.doi.org/10.1521/jscp.2007.26.4.505>
- Mond, J., Mitchison, D., Latner, J., Hay, P., Owen, C., & Rodgers, B. (2013). Quality of life impairment associated with body dissatisfaction in a general population sample of women. *BMC Public Health*, 13, 920. <http://dx.doi.org/10.1186/1471-2458-13-920>
- Mond, J., van den Berg, P., Boutelle, K., Hannan, P., & Neumark-Sztainer, D. (2011). Obesity, body dissatisfaction, and emotional well-being in early and late adolescence: Findings from the project EAT study. *Journal of Adolescent Health*, 48, 373–378. <http://dx.doi.org/10.1016/j.jadohealth.2010.07.022>
- Mutale, G. J., Dunn, A. K., Stiller, J., & Larkin, R. (2016). Development of a Body Dissatisfaction Scale assessment tool. *The New School Psychology Bulletin*, 13, 47–57.
- Myers, L., & Sirois, M. J. (2006). Spearman correlation coefficients, differences between. *Encyclopedia of Statistical Sciences*. Advance online publication. <http://dx.doi.org/10.1002/0471667196.ess5050.pub2>
- Novella, J., Gosselin, J. T., & Danowski, D. (2015). One size doesn’t fit all: New continua of figure drawings and their relation to ideal body image. *Journal of American College Health*, 63, 353–360. <http://dx.doi.org/10.1080/07448481.2015.1040410>
- Olivardia, R. (2001). Mirror, mirror on the wall, who’s the largest of them all? The features and phenomenology of muscle dysmorphia. *Harvard Review of Psychiatry*, 9, 254–259. <http://dx.doi.org/10.1080/10673220127900>
- Olivardia, R., Pope, H. G., Jr., Borowiecki, J. J., III & Cohane, G. H. (2004). Biceps and body image: The relationship between muscularity and self-esteem, depression, and eating disorder symptoms. *Psychology of Men and Masculinity*, 5, 112–120. <http://dx.doi.org/10.1037/1524-9220.5.2.112>
- Paxton, S. J., Neumark-Sztainer, D., Hannan, P. J., & Eisenberg, M. E. (2006). Body dissatisfaction prospectively predicts depressive mood and low self-esteem in adolescent girls and boys. *Journal of Clinical Child and Adolescent Psychology*, 35, 539–549. http://dx.doi.org/10.1207/s15374424jccp3504_5
- Peters, R., Copeland, J., & Dillon, P. (1999). Anabolic-androgenic steroids: User characteristics, motivations, and deterrents. *Psychology of Addictive Behaviors*, 13, 232–242. <http://dx.doi.org/10.1037/0893-164X.13.3.232>
- Pope, H. G., Jr., Gruber, A. J., Mangweth, B., Bureau, B., deCol, C., Jouvent, R., & Hudson, J. I. (2000). Body image perception among men in three countries. *The American Journal of Psychiatry*, 157, 1297–1301. <http://dx.doi.org/10.1176/appi.ajp.157.8.1297>
- Pope, H. G., Phillips, K. A., & Olivardia, R. (2000). *The Adonis complex: The secret crisis of male body image obsession*. New York, NY: Free Press.
- Reas, D. L., Øverås, M., & Rø, O. (2012). Norms for the Eating Disorder Examination Questionnaire (EDE-Q) among high school and university men. *Eating Disorders: The Journal of Treatment and Prevention*, 20, 437–443. <http://dx.doi.org/10.1080/10640266.2012.715523>
- Rhodes, G., & Simmons, L. W. (2007). Symmetry, attractiveness and sexual selection. *Oxford Handbook of Evolutionary Psychology*. Advance online publication. <http://dx.doi.org/10.1093/oxfordhb/9780198568308.013.0024>
- Ryan, T. A., Morrison, T. G., Roddy, S., & McCutcheon, J. (2011). Psychometric properties of the Revised Male Body Attitudes Scale among Irish men. *Body Image*, 8, 64–69. <http://dx.doi.org/10.1016/j.bodyim.2010.10.004>
- Sánchez, F. J., Greenberg, S. T., Liu, W. M., & Vilain, E. (2009). Reported effects of masculine ideals on gay men. *Psychology of Men and Masculinity*, 10, 73–87. <http://dx.doi.org/10.1037/a0013513>
- Strubel, J., & Petrie, T. A. (2017). Love me Tinder: Body image and psychosocial functioning among men and women. *Body Image*, 21, 34–38. <http://dx.doi.org/10.1016/j.bodyim.2017.02.006>
- Swami, V., Salem, N., Furnham, A., & Tovée, M. J. (2008). Initial examination of the validity and reliability of the female photographic figure rating scale for body image assessment. *Personality and Individ-*

- ual Differences, 44, 1752–1761. <http://dx.doi.org/10.1016/j.paid.2008.02.002>
- Tiggemann, M., Martins, Y., & Kirkbride, A. (2007). Oh to be lean and muscular: Body image ideals in gay and heterosexual men. *Psychology of Men and Masculinity*, 8, 15–24. <http://dx.doi.org/10.1037/1524-9220.8.1.15>
- Tovée, M. J., Tasker, K., & Benson, P. J. (2000). Is symmetry a visual cue to attractiveness in the human female body? *Evolution and Human Behavior*, 21, 191–200. [http://dx.doi.org/10.1016/S1090-5138\(00\)00040-4](http://dx.doi.org/10.1016/S1090-5138(00)00040-4)
- Wardle, J., & Cooke, L. (2005). The impact of obesity on psychological well-being. *Best Practice and Research Clinical Endocrinology and Metabolism: Clinical and Experimental*, 19, 421–440. <http://dx.doi.org/10.1016/j.beem.2005.04.006>
- Watkins, J. A., Christie, C., & Chally, P. (2008). Relationship between body image and body mass index in college men. *Journal of American College Health*, 57, 95–100. <http://dx.doi.org/10.3200/JACH.57.1.95-100>
- Wright, S., Grogan, S., & Hunter, G. (2000). Motivations for anabolic steroid use among bodybuilders. *Journal of Health Psychology*, 5, 566–571. <http://dx.doi.org/10.1177/135910530000500413>
- Yelland, C., & Tiggemann, M. (2003). Muscularity and the gay ideal: Body dissatisfaction and disordered eating in homosexual men. *Eating Behaviors*, 4, 107–116. [http://dx.doi.org/10.1016/S1471-0153\(03\)00014-X](http://dx.doi.org/10.1016/S1471-0153(03)00014-X)

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